

REAL STEEL

EN 2533 Robotics Design and Competition

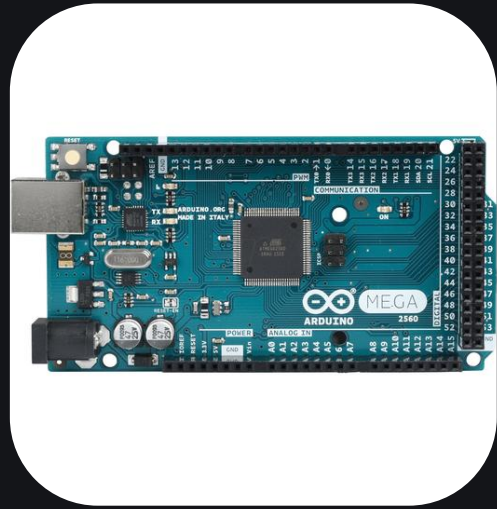
230689A - Weerakoon A.H.T.M.

230138K - Dhananjaya K.T.G.T.N.

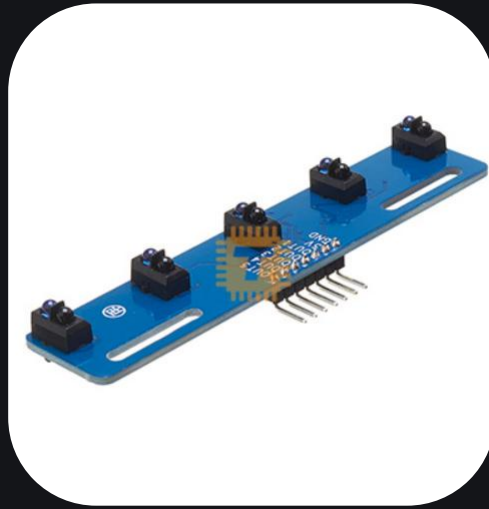
230155J - Dissanayaka D.M.D.P.

230183R - Fernando L.T.J.

COMPONENTS SELECTION



**Arduino
mega 2560**



IR array



Colour sensor



Sharp IR



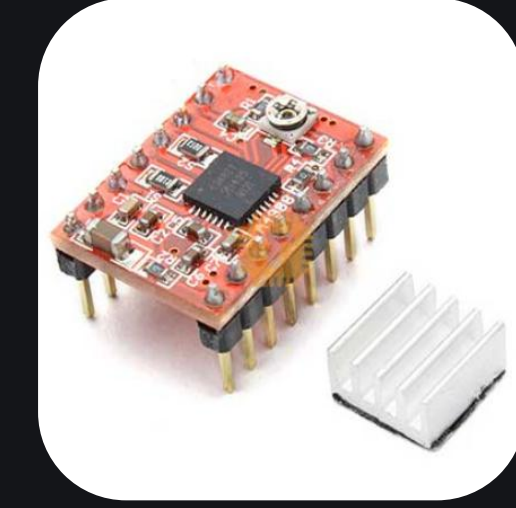
Servo



**Stepper
Motor**



Wheels



**Stepper
Motor Driver**



**Stepper
Motor for
arm**



**Stepper
Motor Driver
for arm**



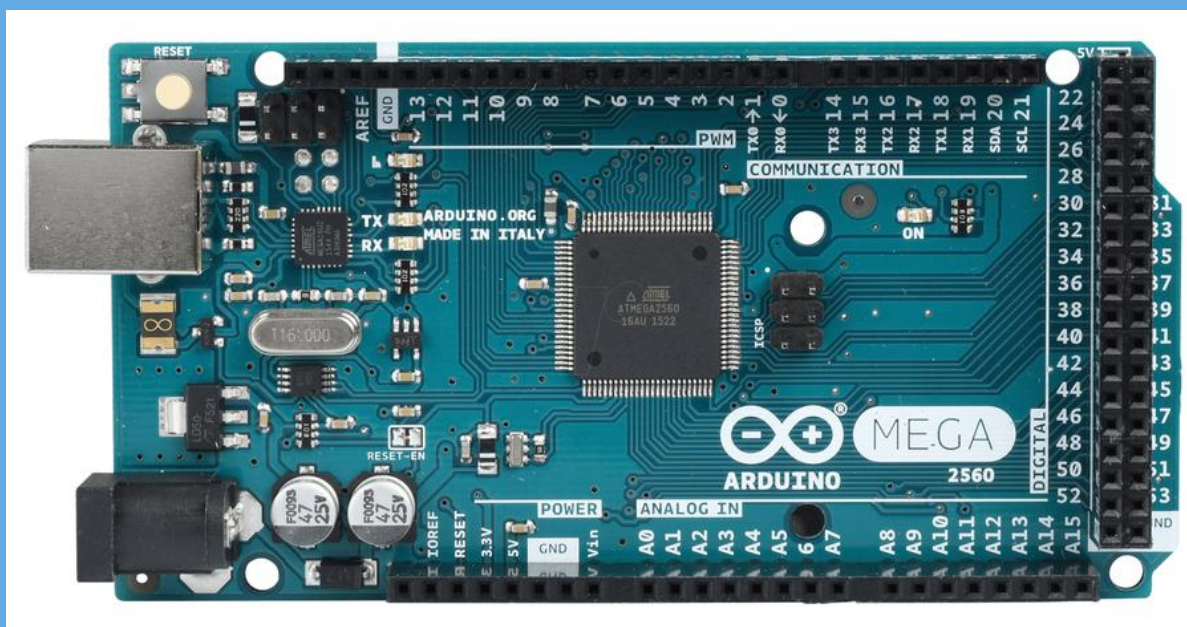
**Acrylic sheets
3mm²**

Microcontroller

Arduino mega

2560

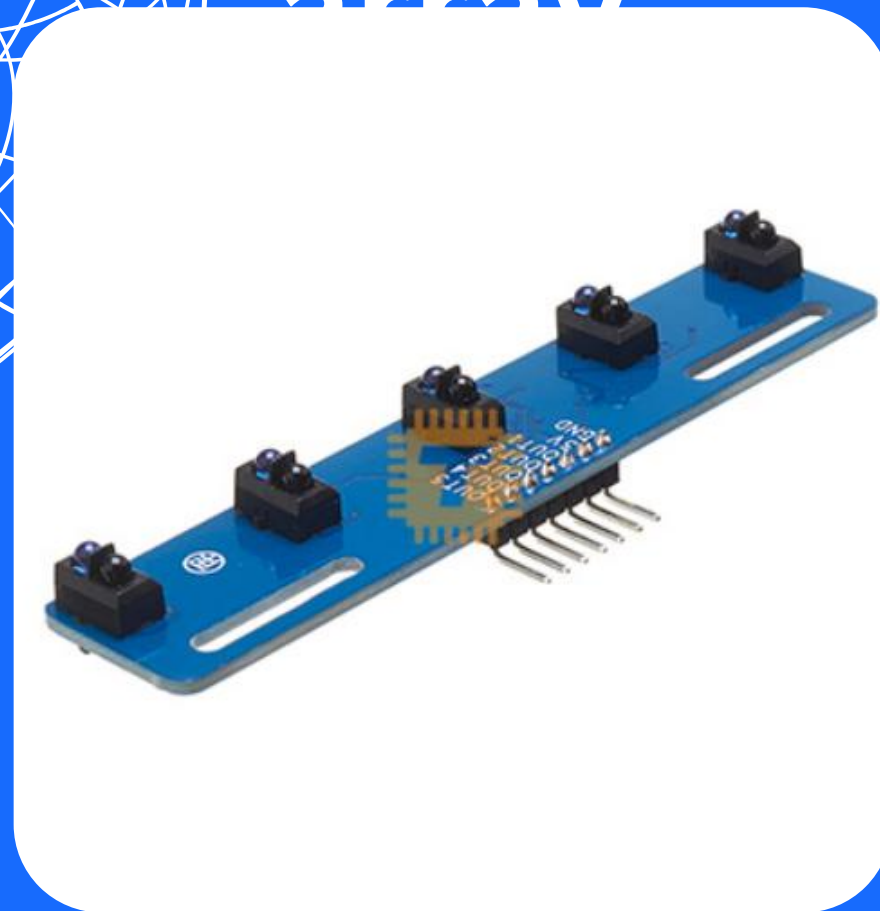
- Microcontroller: ATmega2560
- Operating Voltage: 5V
- Input Voltage (limits): 6–20V
- Clock Speed: 16 MHz
- Flash Memory: 256 KB (of which 8 KB used by bootloader)
- SRAM: 8 KB
- EEPROM: 4 KB
- Digital I/O Pins: 54 (of which 15 can provide PWM output)
- Analog Input Pins: 16
- Analog Output: PWM on 15 pins (no true DAC)
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Max DC Current for 5V Pin: 500 mA (limited by onboard regulator & USB)
- UARTs (Serial Ports): 4 (Serial, Serial1, Serial2, Serial3)
- I²C (TWI): Yes
- SPI: Yes
- USB: Type-B connector (for programming and communication)
- ICSP header: Available



Sensors

IR

array



- 5 IR Line hunting sensor module

- Distance between two

: 0.25

Distance sensor



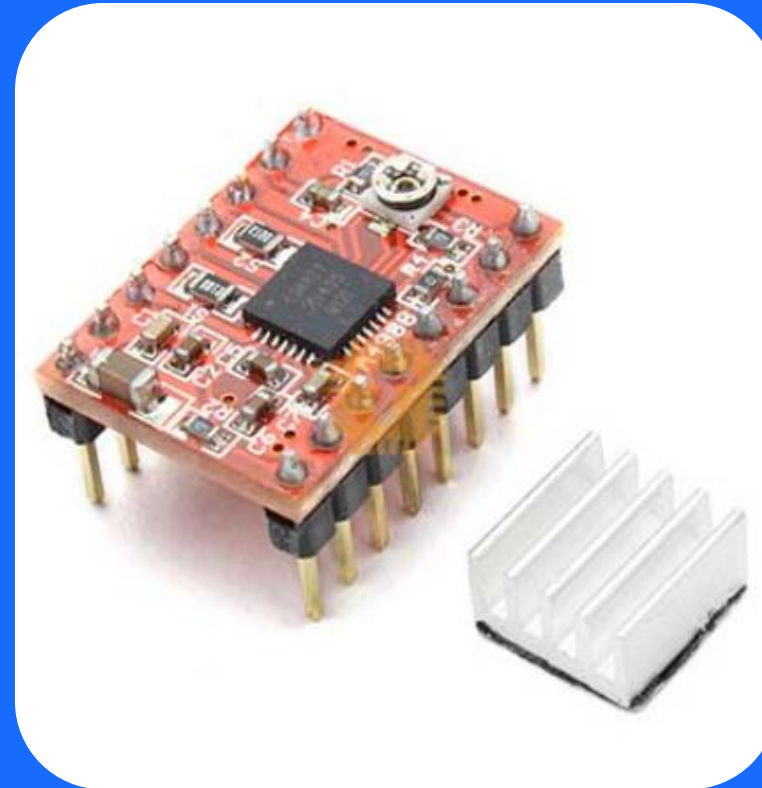
- SHARP GP2Y0A21YK0F Infrared IR Sensor (10- 80cm)
- Narrow and high accurate measuring.

Colour sensor



- TCS34725 RGB Colour Recognition Sensor
- I2C protocol, high accuracy.

Motor, Wheels and Motor Drivers



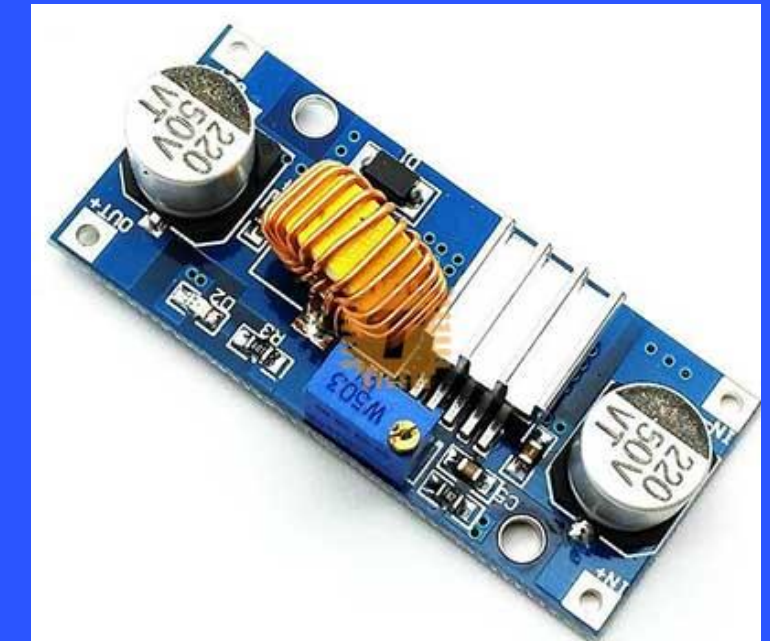
- Nema 17 [Stepper Motor 17HS4401](#).
- 12V 1.7A -- 40Ncm torque.
- Precise control of motion.

- A4988 Stepper Motor Driver
- Operating voltage: 8-35V
- Current per phase: 1A-2A

- Rubber, high-grip
- 85mm Diameter
- Tyre width: 31mm.

Power Battery

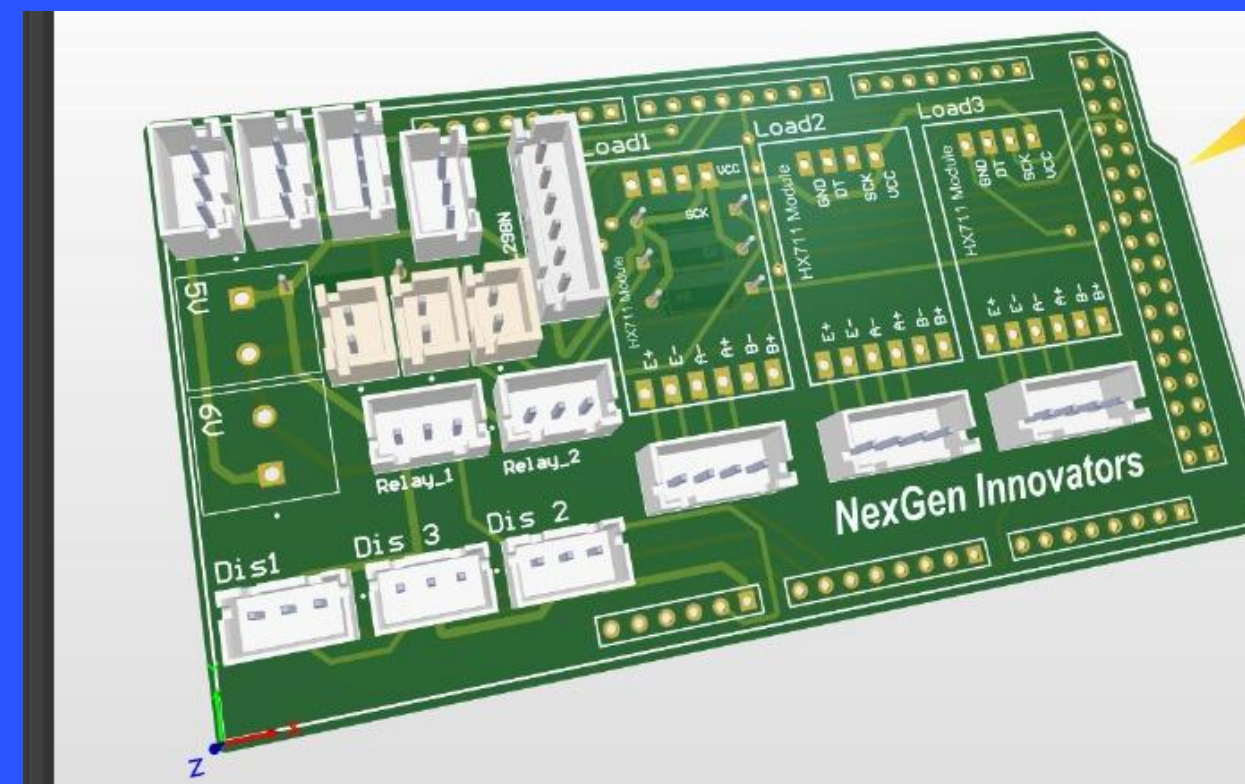
- Li Ion 18650 3.7V 3200mAh 4 cells --- 14.4V
- Step down buck converters to get 12V, 5V and 3.3V seperately.
- This ensures the constant voltage.



PCB

design

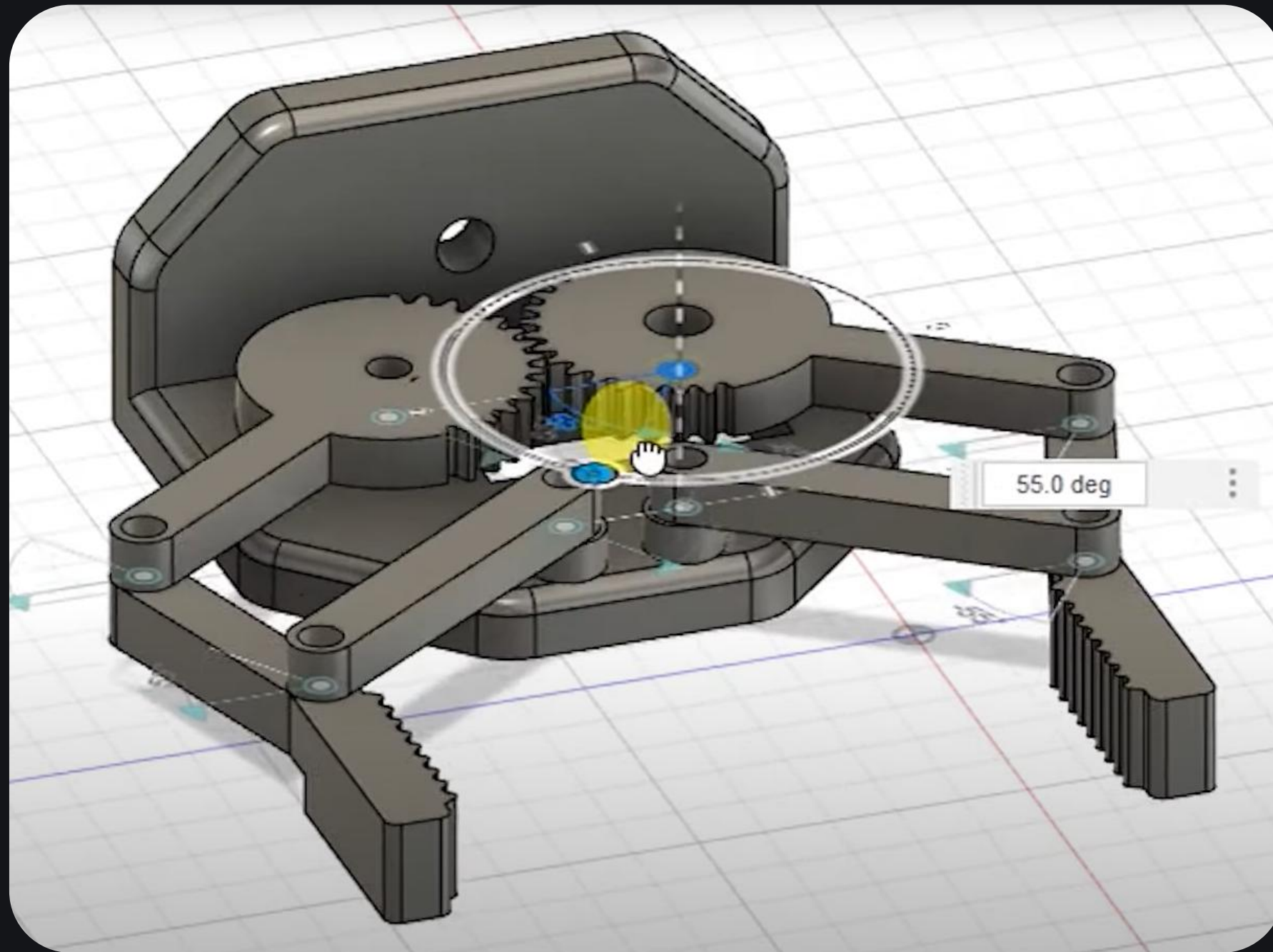
- Custom PCB is designed for better connection of power and I/O.



Sample Design as a sheild for arduino

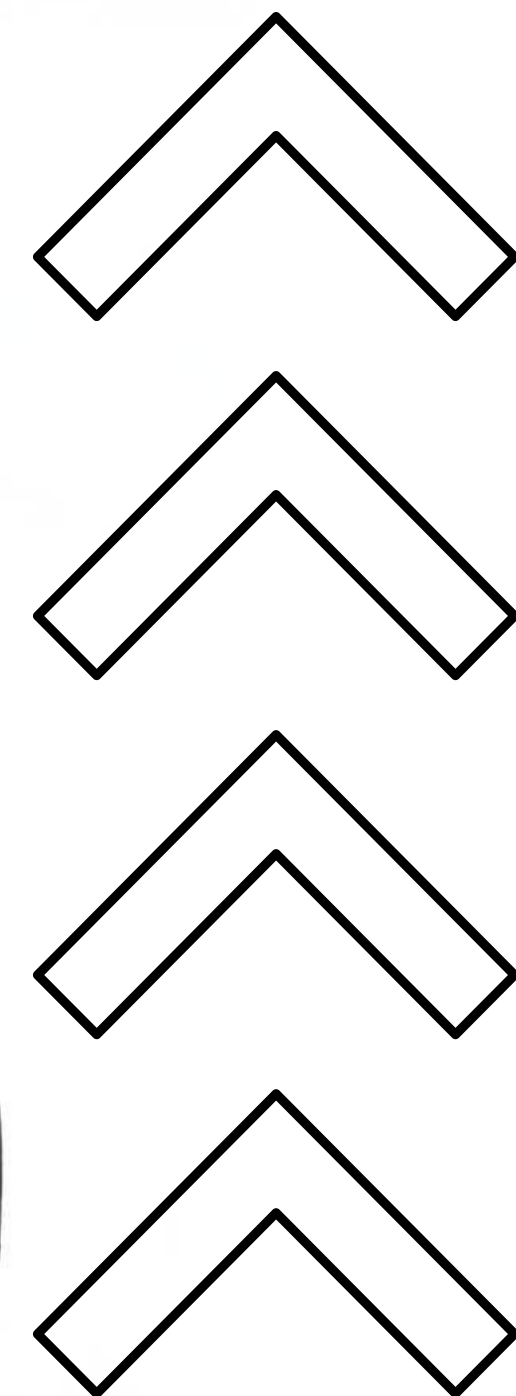


Proposed Arm design for boxes

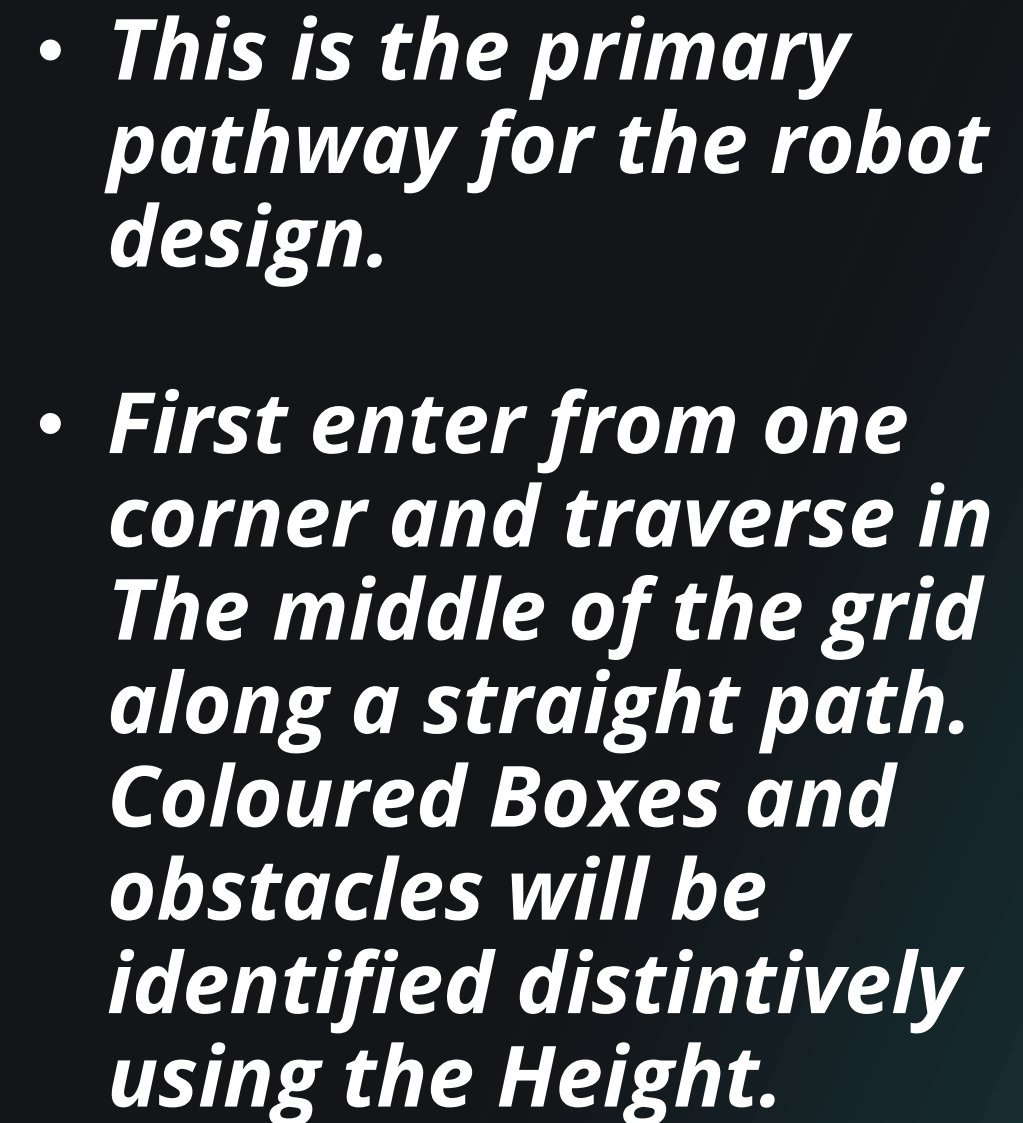


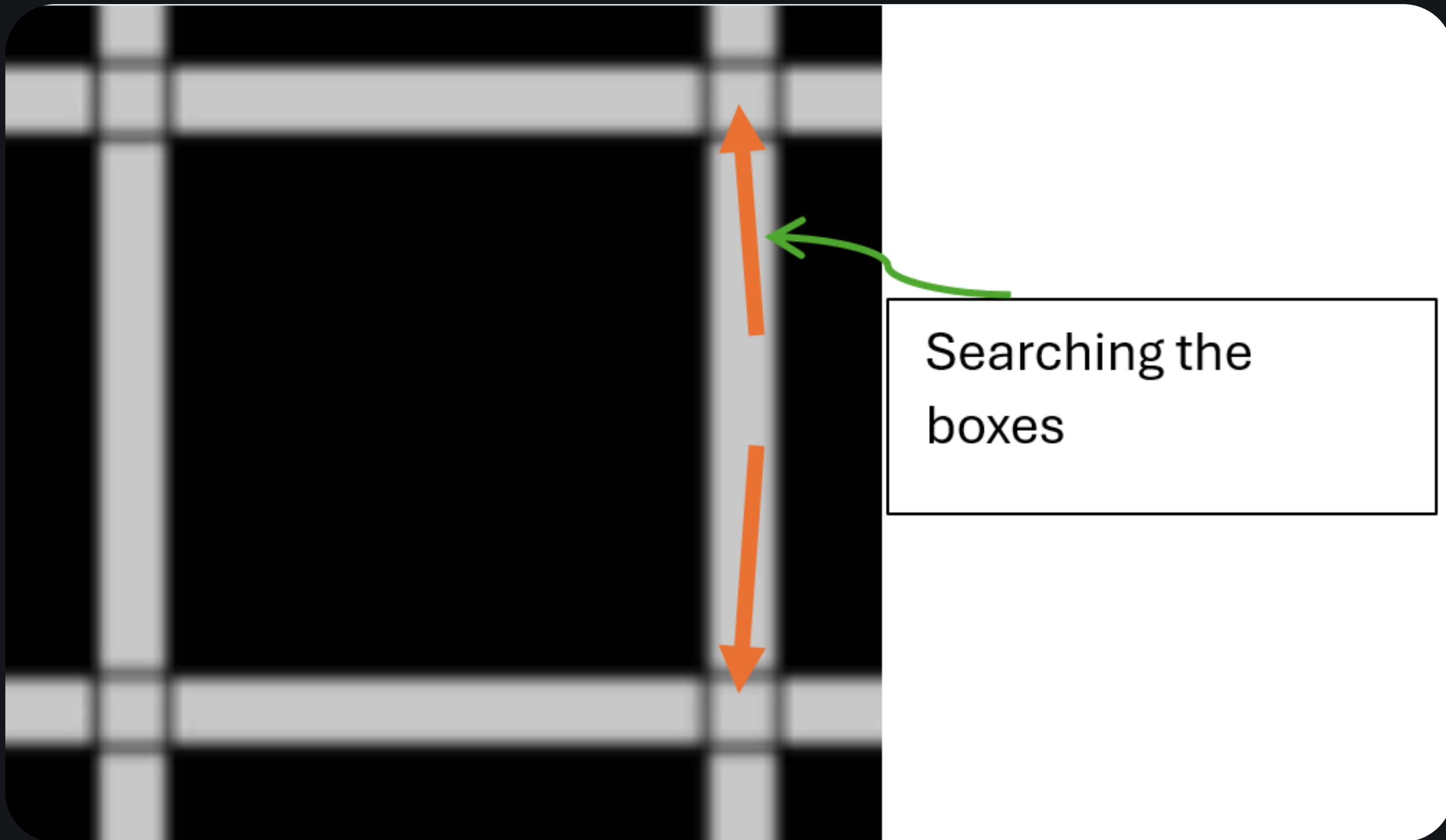
- Gripping the boxes using a single servo mechanism.
- Arm lifting using a stepper motor. (28BYJ-48 5V stepper).

Picture taken from
internet



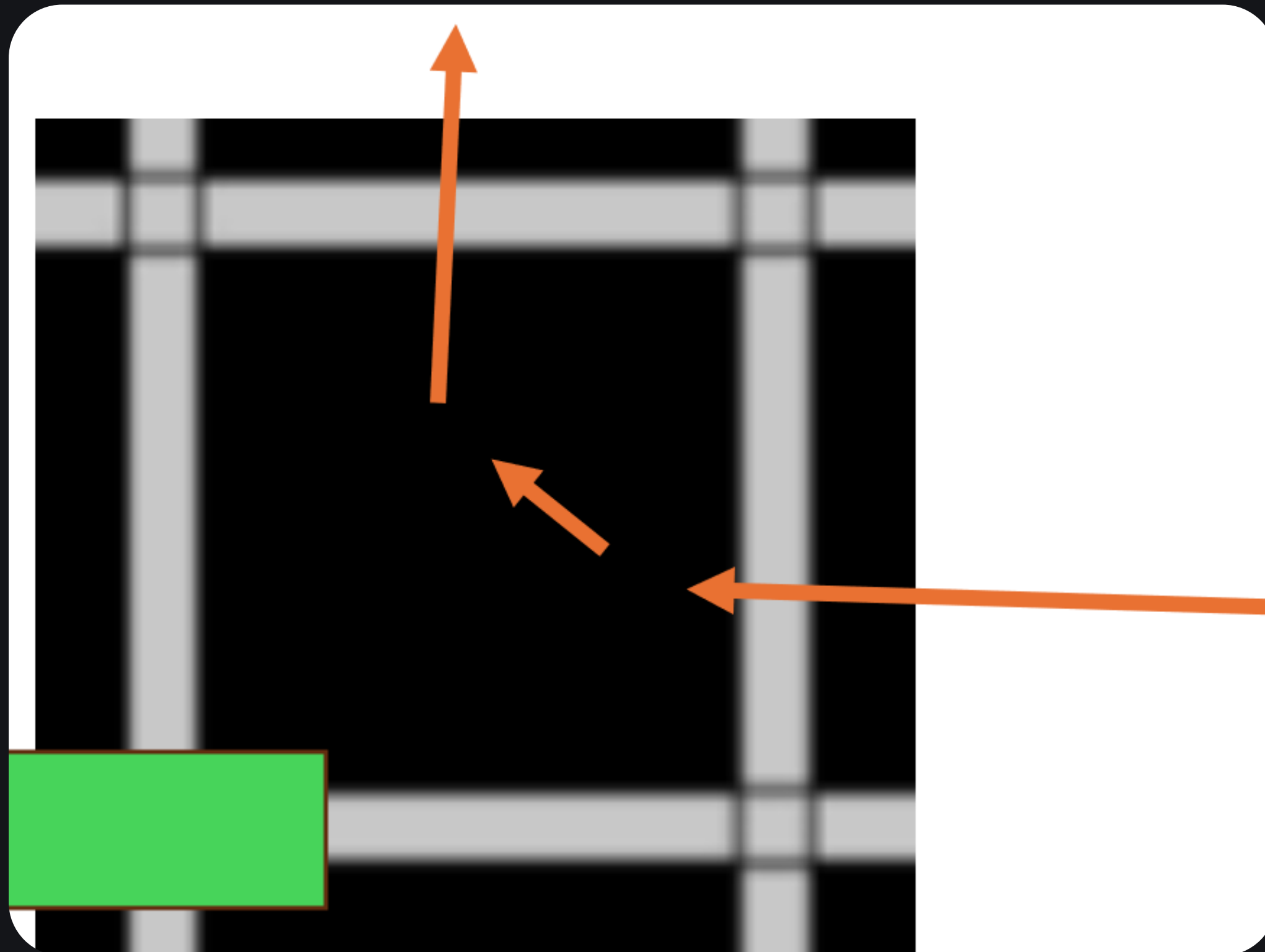
THE LABYRINTH OF SHADOWS (GRID SOLVING)





- *Finding a coloured box in the middle aligned with a white line.*
- *Robot will follow the line and grab the box. Come back to middle of the lines and get to a corner row of the grid. Continue to go until it reaches the matching-coloured area to drop the box.*
- *Robot will count the white lines crossed to get the position of robot in the grid.*

Box detection

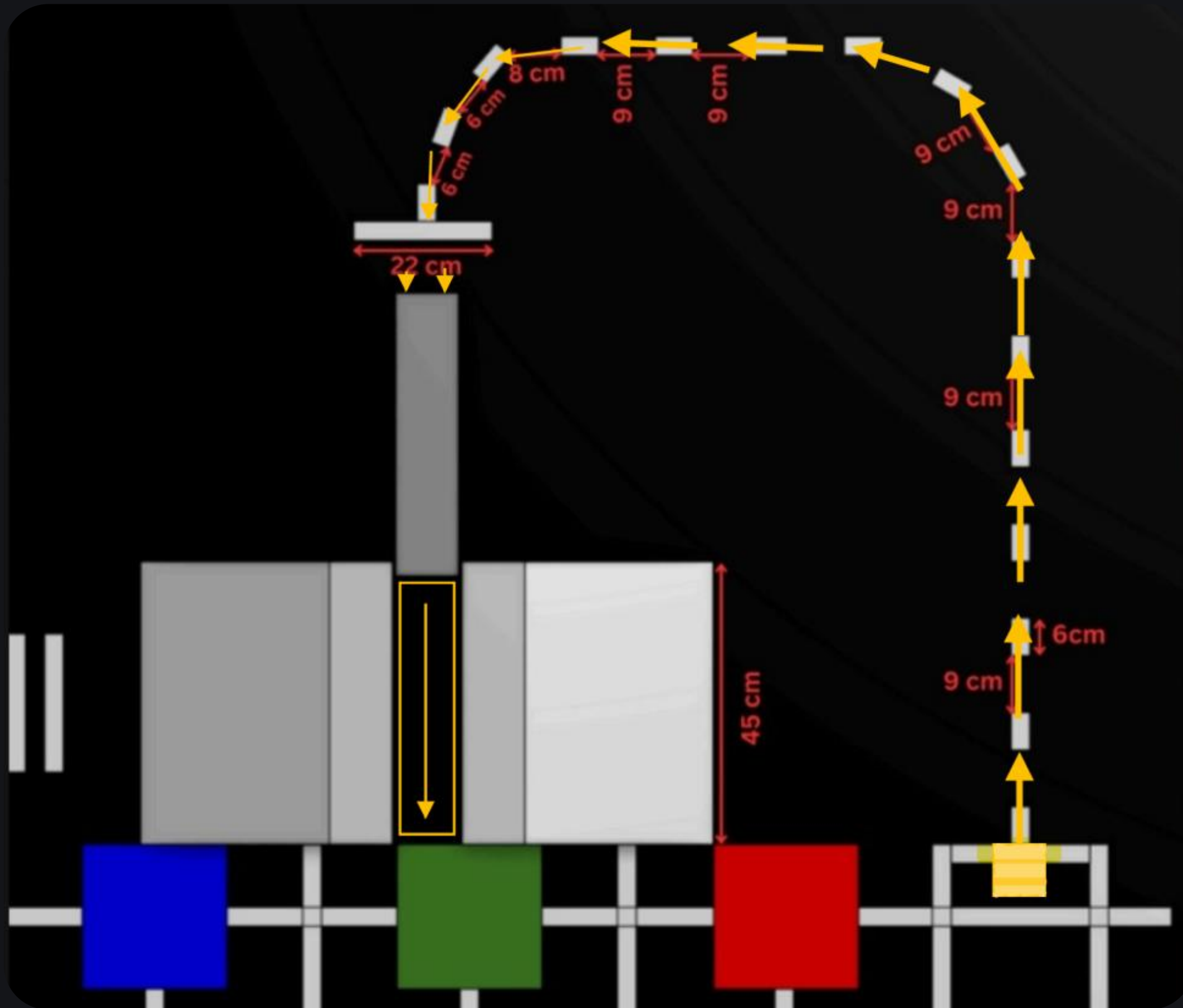


In the corners

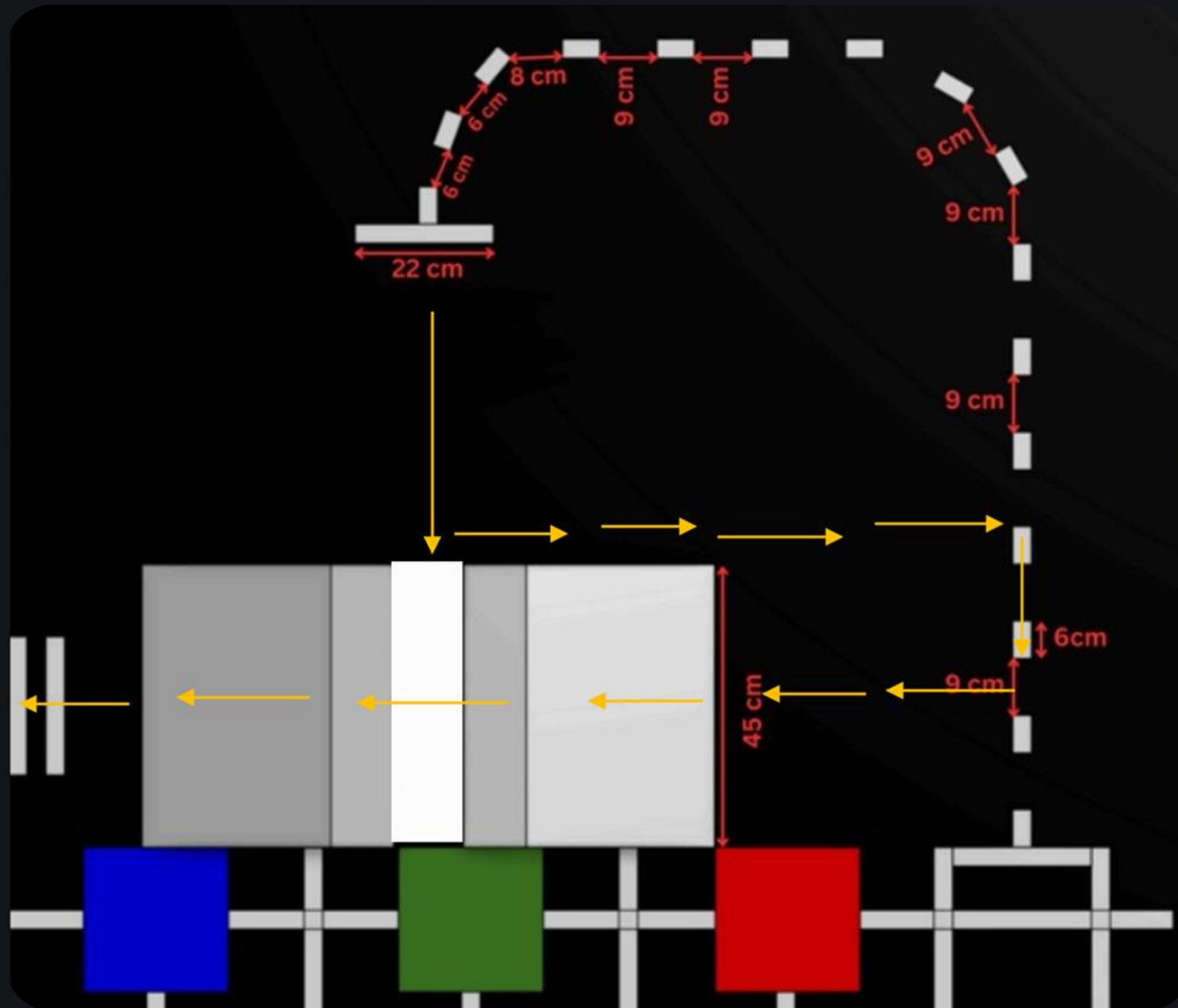
- ***Finding a coloured box in the corners of a single cell.***
- ***In the middle the robot will turn and search for the box using distance sensors. Then it will come to the forward and follow the white line to reach the box. After grabbing the box, the robot will come back to the middle and follow it until the coloured box.***

TASK 02

THE RAMP OF TRIALS



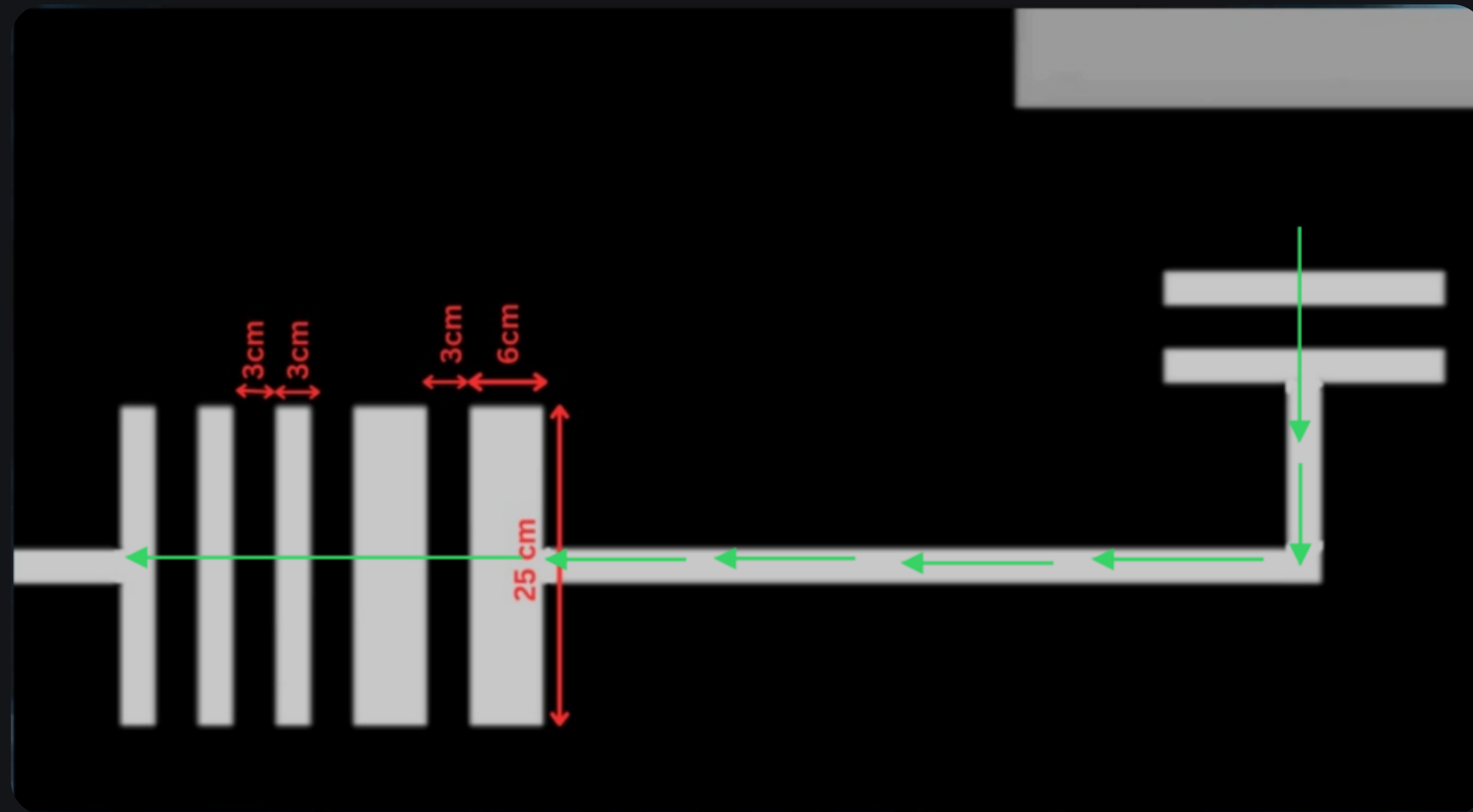
- *Follow the line until reaches to the missing part of the ramp.*
- *Move the missing part so that the ramp completes.*



- *Thereafter go back to the dash line and comes to the starting point of the ramp*

TASK 03

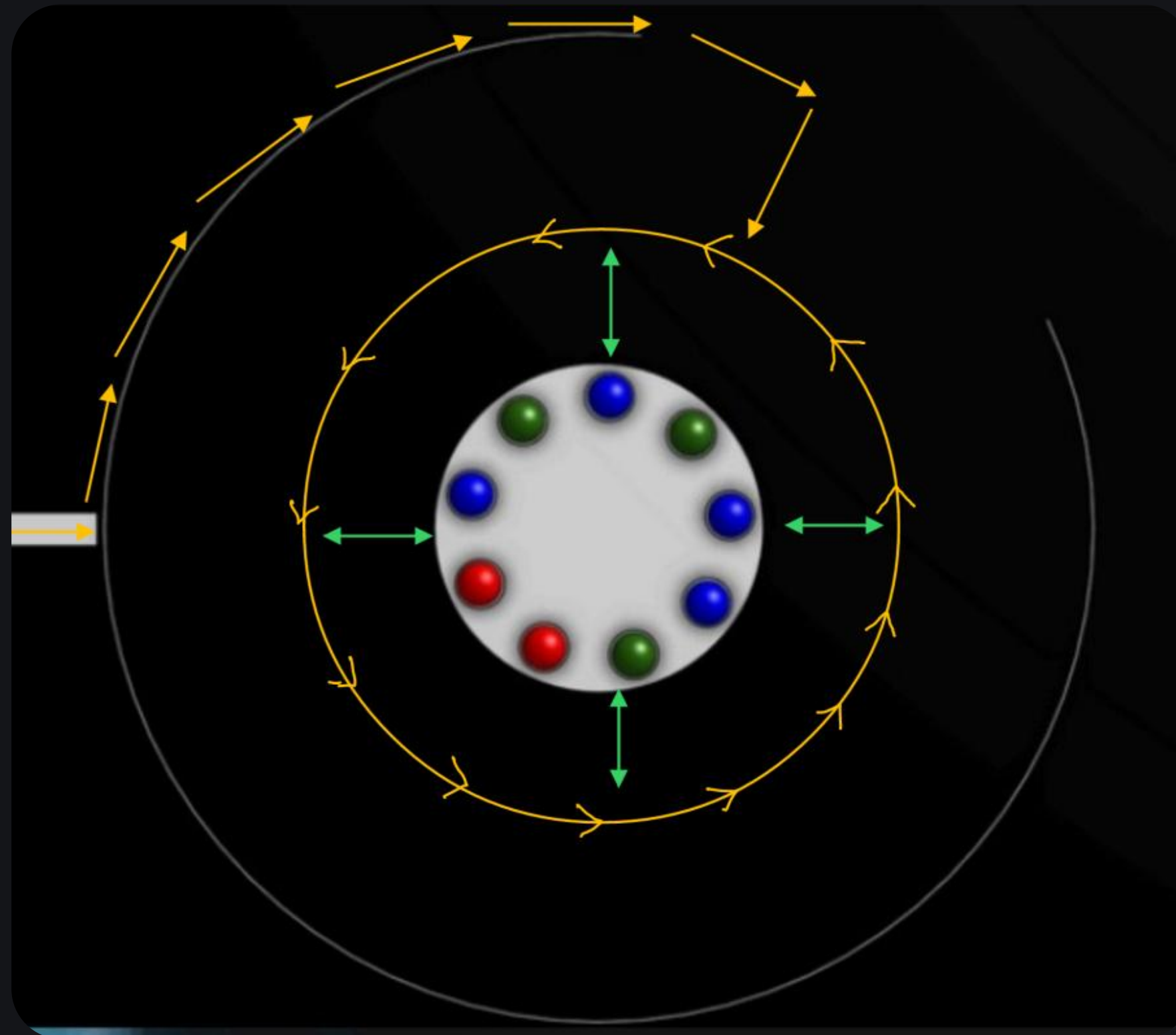
THE ORACLE OF RUNES(BARCODE READING)



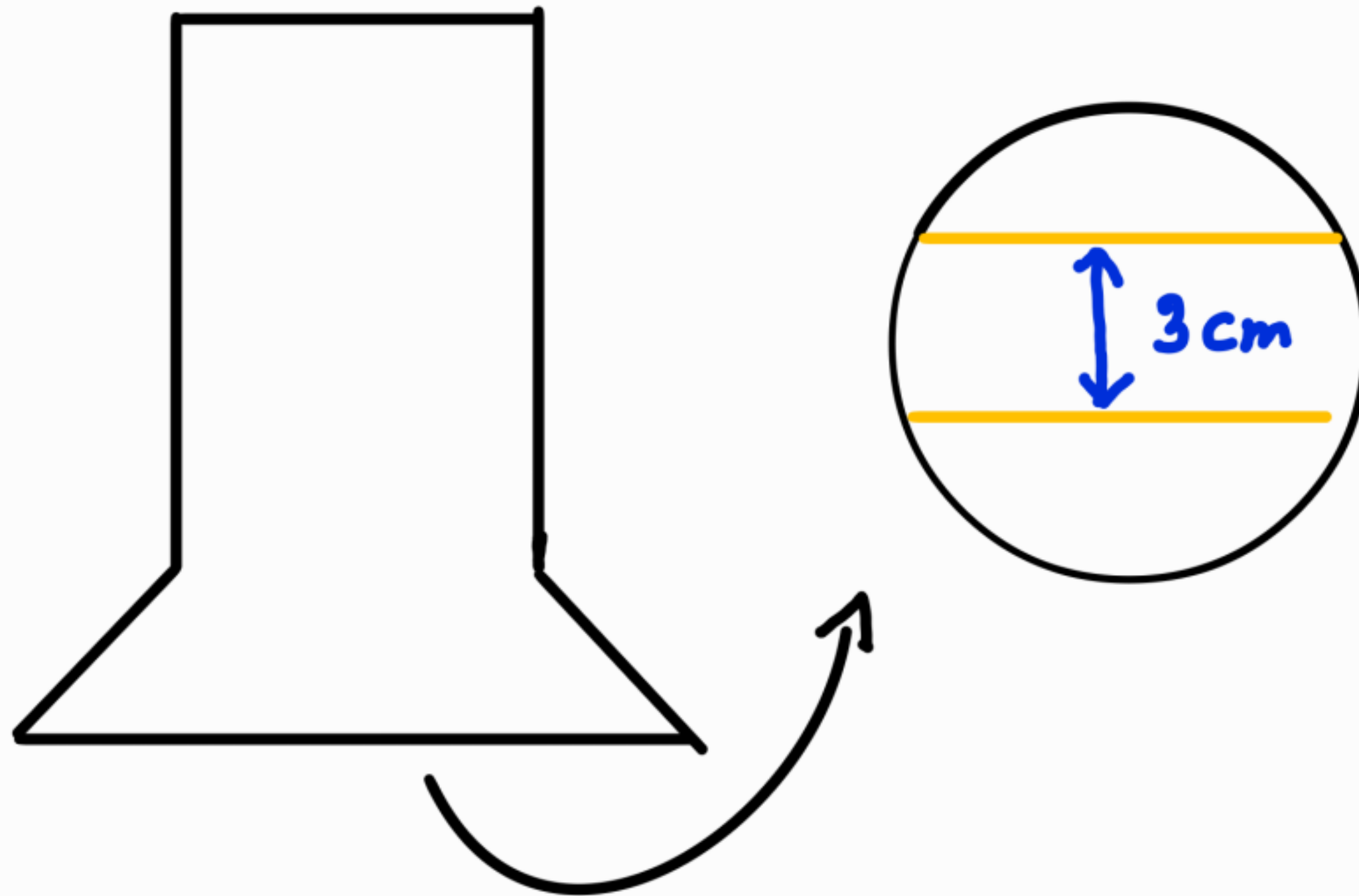
- *Move through the Bar code.*
- *While moving measure the length of the lines and compute the integer value.*

TASK 04

THE CIRCULAR FORTRESS(WALL FOLLOWING AND BALL COLLECTION)



- *Follow the wall and find the open area.*
- *Enters to the circle and move around center stage by keeping a constant distance from the wall.*
- *While traversing position of the balls will be identified using the distance sensors.*



- *This is used to pick the balls.*

THE WALL OF WHISPERS & PENDULUM OF DOOM

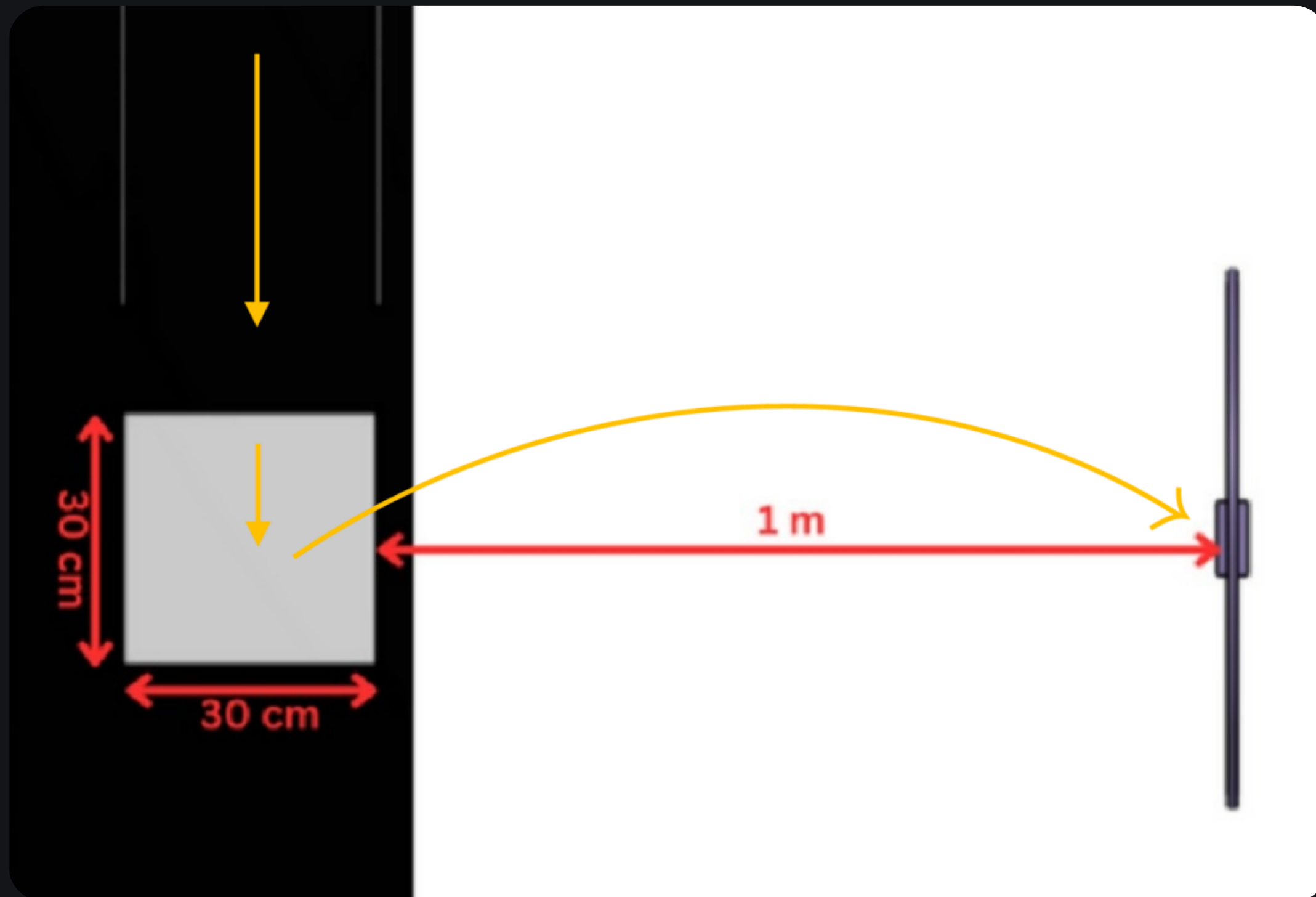
-
- A diagram showing a robot's path in a hallway. The path is a closed loop starting from a blue dot at the bottom center. The path consists of several segments: a vertical segment up (25cm), a horizontal segment right, a diagonal segment down-right, a vertical segment up, a diagonal segment up-left, a horizontal segment left, a diagonal segment up-left, a horizontal segment right (30cm), and a vertical segment down. Yellow arrows indicate the direction of travel along the path.



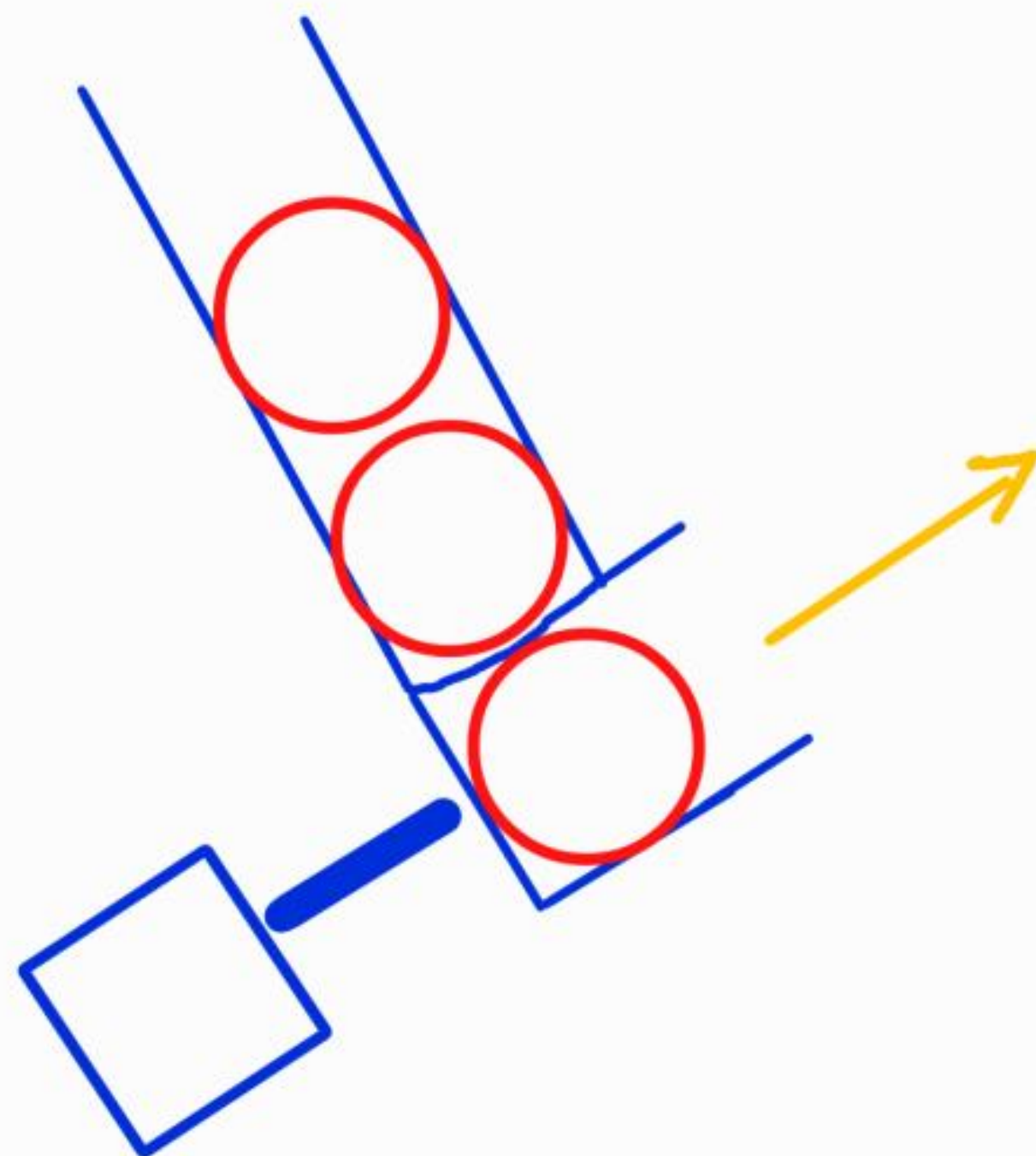
- *Follow the walls and complete the hidden task.*
- *pass through the gate and reach to the final task.*

TASK 06

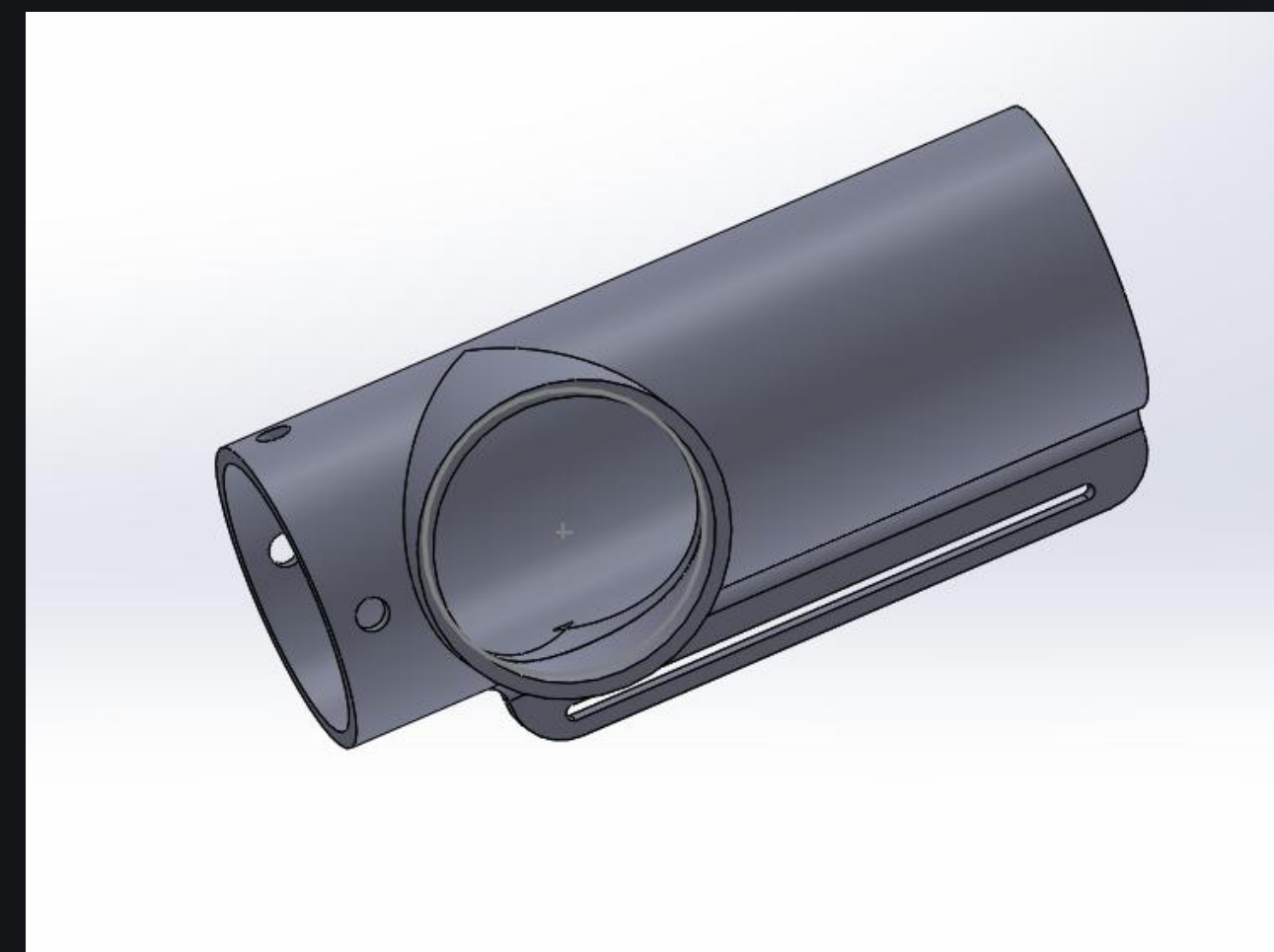
THE LINE OF DESTINY & THE RING OF TRIUMPH(SHOOTING TASK)



- *Reach to the white square and place the robot facing at the goal.*
- *There after aim the shooter and throw the balls.*



- ***This is used to throw the balls .***



Solidworks Design

Robot Design Roadmap

	Aug 29	Sept 9	Sept 14	Sept 17	Sept 26	Sept 30	Oct 4	Oct 7
Task Solution Plannina	DONE							
Buuina Components		DONE						
Sensor Checkina			DONE					
Arm Desian/Motor Testina				DONE				
chasis Desian					To Do			
Line follower buildina					To Do			
Arm Desina and Testina						To Do		
Ball Pickina Task Desian							To Do	

Robot Design Roadmap

	Oct 7	Oct 11	Oct 14	Oct 18
Ball Shooter Desian/ Proorammina task 1.2	To Do			
Ball Shooter Testina		To Do		
Finalizina Hardware of Robot			To Do	
Proorammina			To Do	
Testina			To Do	
Finalizina the desian				To Do

COST ANALYSIS

<i>Jumper wires</i>	-	<i>Rs. 980.00</i>
<i>Batteries</i>	-	<i>Rs. 1,060.00</i>
<i>Voltage convertors</i>	-	<i>Rs. 900.00</i>
<i>Spacers</i>	-	<i>Rs. 400.00</i>
<i>Motors</i>	-	<i>Rs. 1,330.00</i>
<i>Motor Drivers</i>	-	<i>Rs. 420.00</i>
<i>Colour Sensors</i>	-	<i>Rs. 760.00</i>
<i>Sharp IR</i>	-	<i>Rs. 3,750.00</i>
<i>IR array</i>	-	<i>Rs. 470.00</i>
<i>Other estimated cost</i>	-	<i>Rs.10,000.00</i>
<i>Total Cost</i>	-	<i>Rs. 20,000.00</i>

Q&A

Thank You!

